

	Work Instruction	Doc. Revision	00
	AXI V810 Motion Waveform Capture for Panasonic Servo Drive	Effective Date	13 Sep 2023

This procedure is applicable to below models:

Machine Model	Server Option		
	S1	S2	S3
V810 / V810i STD			
V810 / V810i XXL			
V810 / V810i S2EX		√	√
V810 / V810i XLT		√	√
V810i XLL		√	√
V810i XLM		√	√
V810i S3		√	√

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	13 Sep 2023	Initial Release	Clement Chong

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Motion Scope Data Capture

1. Prepare a USB Type A Male to Mini USB Type B Male cable (PN: 2530-0242).



Figure 1 USB Type A Male to Mini USB Type B Male cable

2. Connect Type A USB connector to Atom PC.

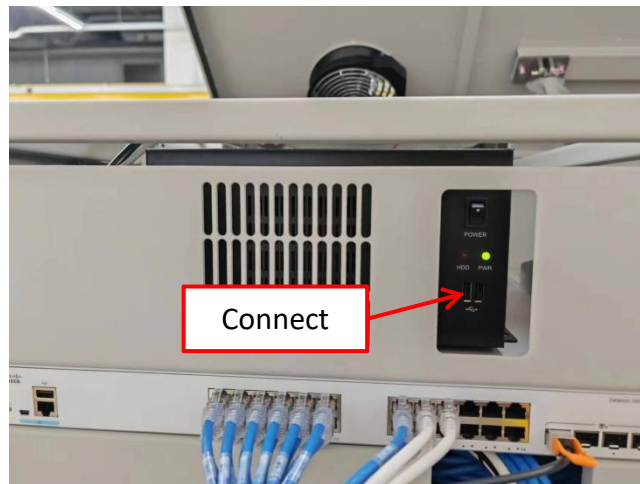


Figure 2 Atom PC Connection

3. For X-axis motion waveform capture, connect Mini USB Type B connector to X-axis Motion Driver X1 port. For Y-axis motion waveform capture, connect Mini USB Type B connector to Y-axis Motion Driver X1 port.

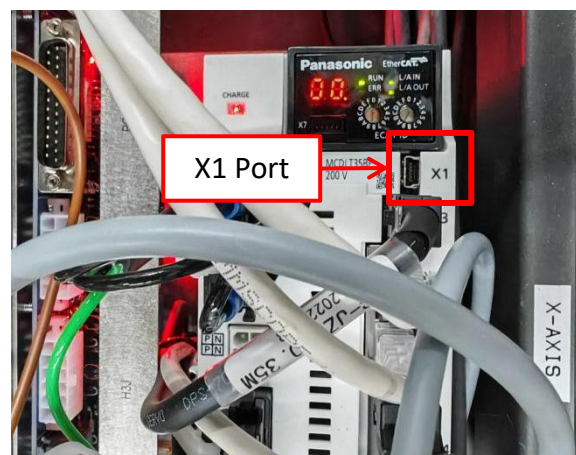


Figure 3 X1 Port

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4. Remote into atom PC via 192.168.128.70.



Figure 4 Remote Desktop Connection (192.168.128.70)

5. Right click on “**PANATERM**” application shortcut at desktop and select “Run as Administrator”. If shortcut is unavailable, the application can be found at path:

“C:\Program Files (x86)\Panasonic Industry\MINAS”



Figure 5 PANATERM application shortcut

6. Ensure that the correct motion driver is detected by referring to the “Drive Nickname”, click “OK” to proceed.

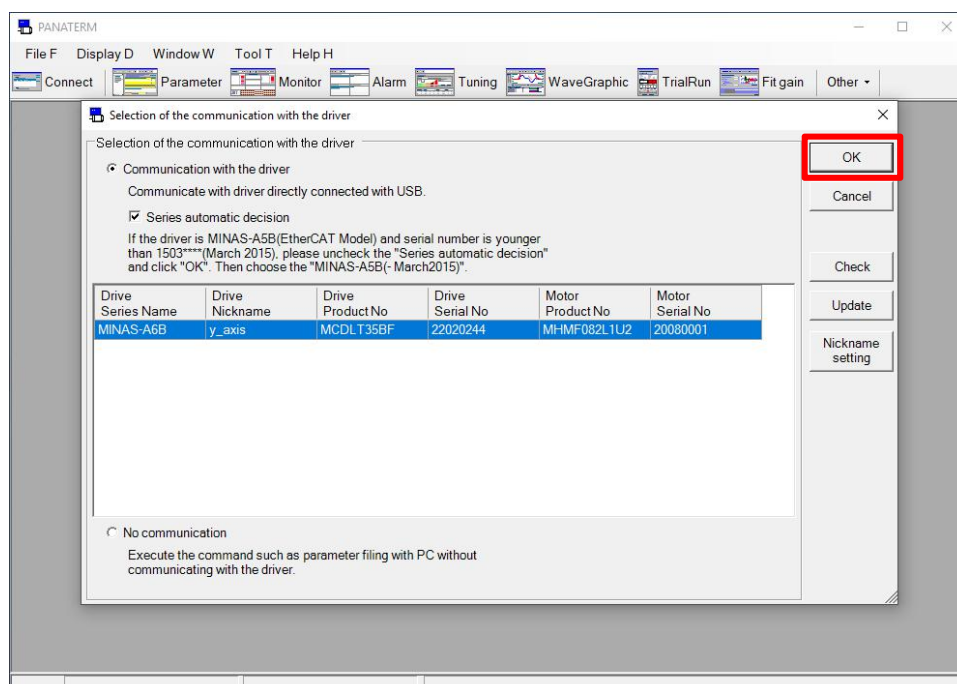


Figure 6 Driver Communication Selection

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7. Go to WaveGraphic.

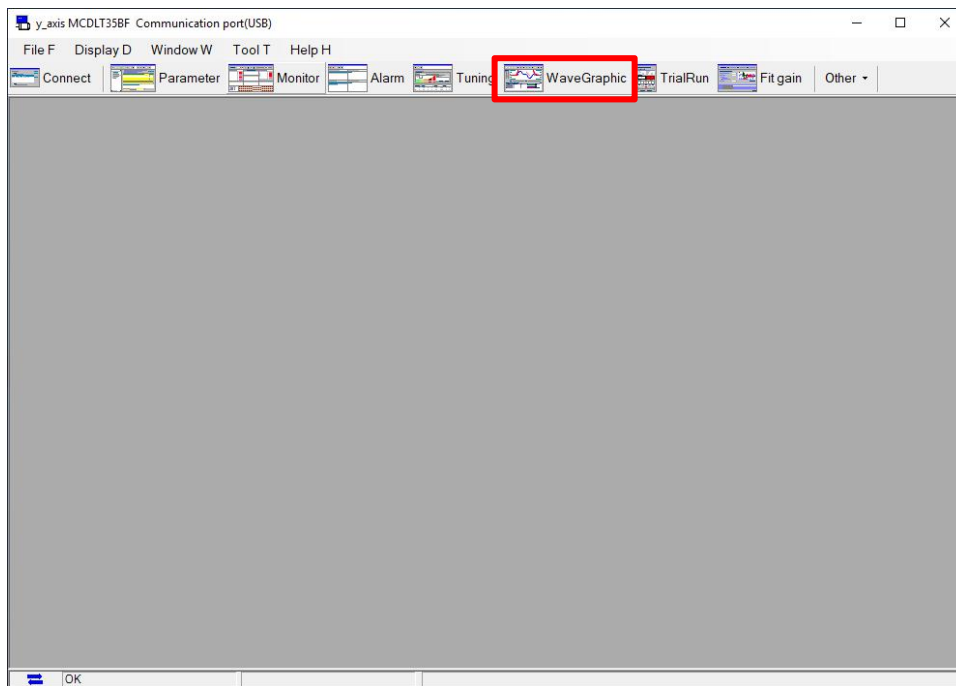


Figure 7 Wave Graphic

8. Double click on any measurement item under the “Measurement” tab to bring up the “Measure Item Select” menu.

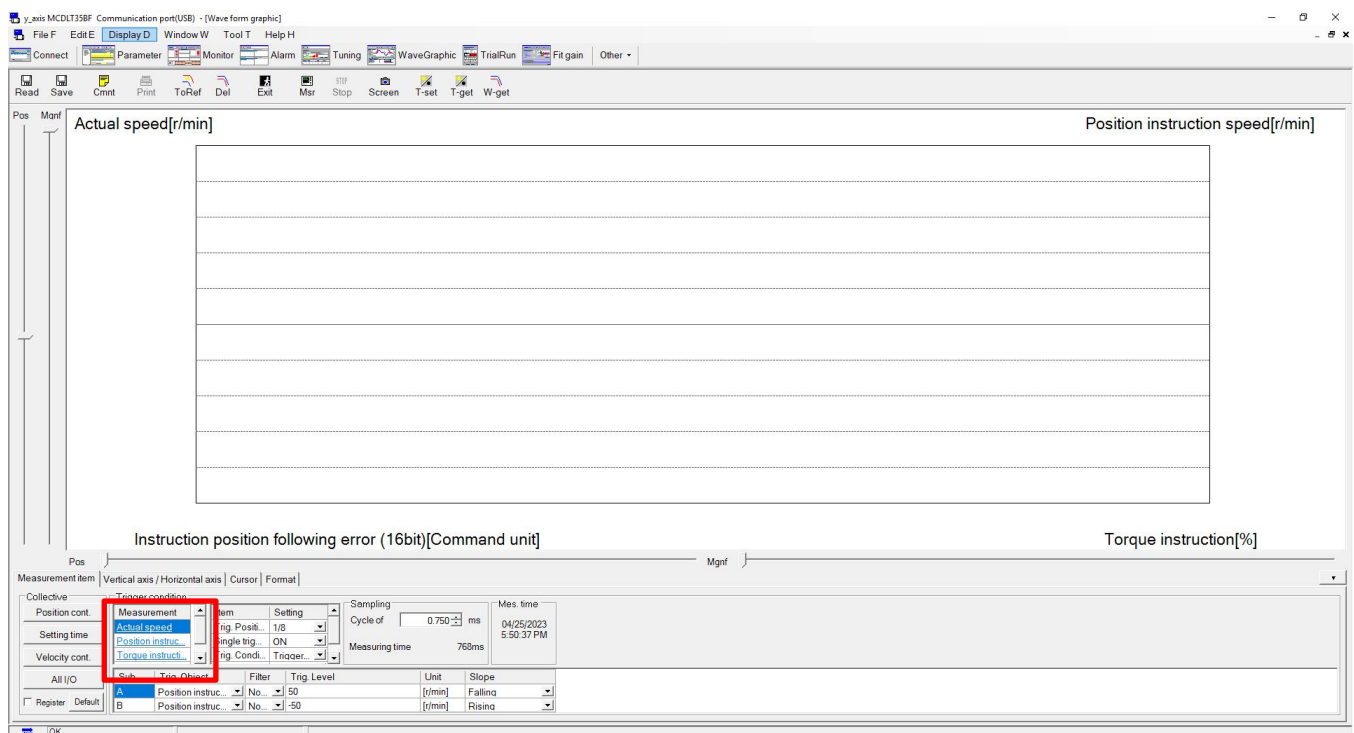


Figure 8 Measurement Tab

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9. Configure the measurement item by referring to Table 1. Click “OK” when done.

Measurement Item	Signal Size
Actual Speed	16
Position Instruction Speed	16
Torque Instruction	16
Full Close Position Following Error	16

Table 1 Measurement Item

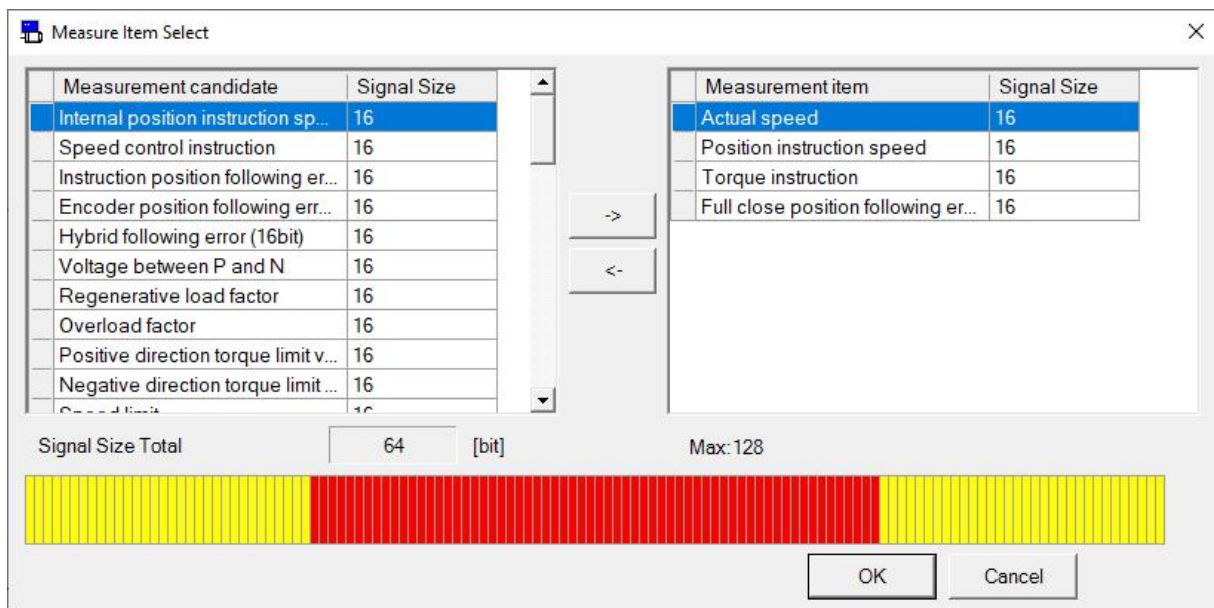


Figure 9 Measurement Item

10. Set sampling cycle time to **30 ms** (Measurement time of 30s).

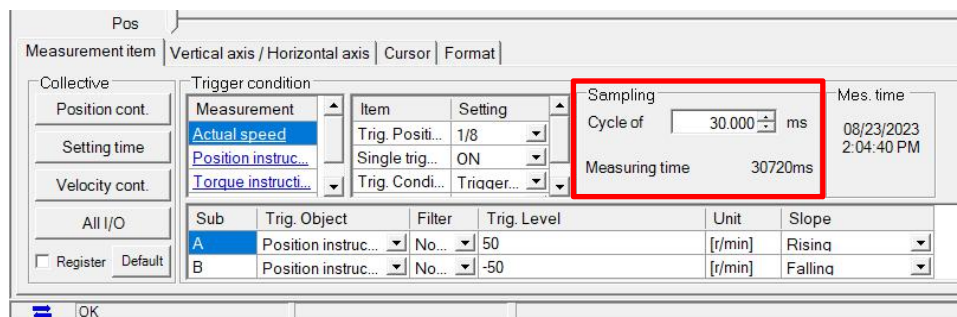


Figure 10 Sampling Time

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11. Configure Trigger Condition by referring to Table 2 and Table 3.

Item	Setting
Trigger position	1/8
Single Trigger	ON
Trigger condition	Triggers A or B

Table 2 Trigger Setting

Sub	Trig. Object	Filter	Trig. Level	Unit	Slope
A	Position instruction speed	No	50	[r/min]	Rising
B	Position instruction speed	No	-50	[r/min]	Falling

Table 3 Trigger Condition

12. Ensure that there is no board inside the machine.

13. At V810 GUI, Go to Services → Subsystem status and Control → Panel Positional → Initialize → Scan Moves → Load Configuration” → load profile → OK

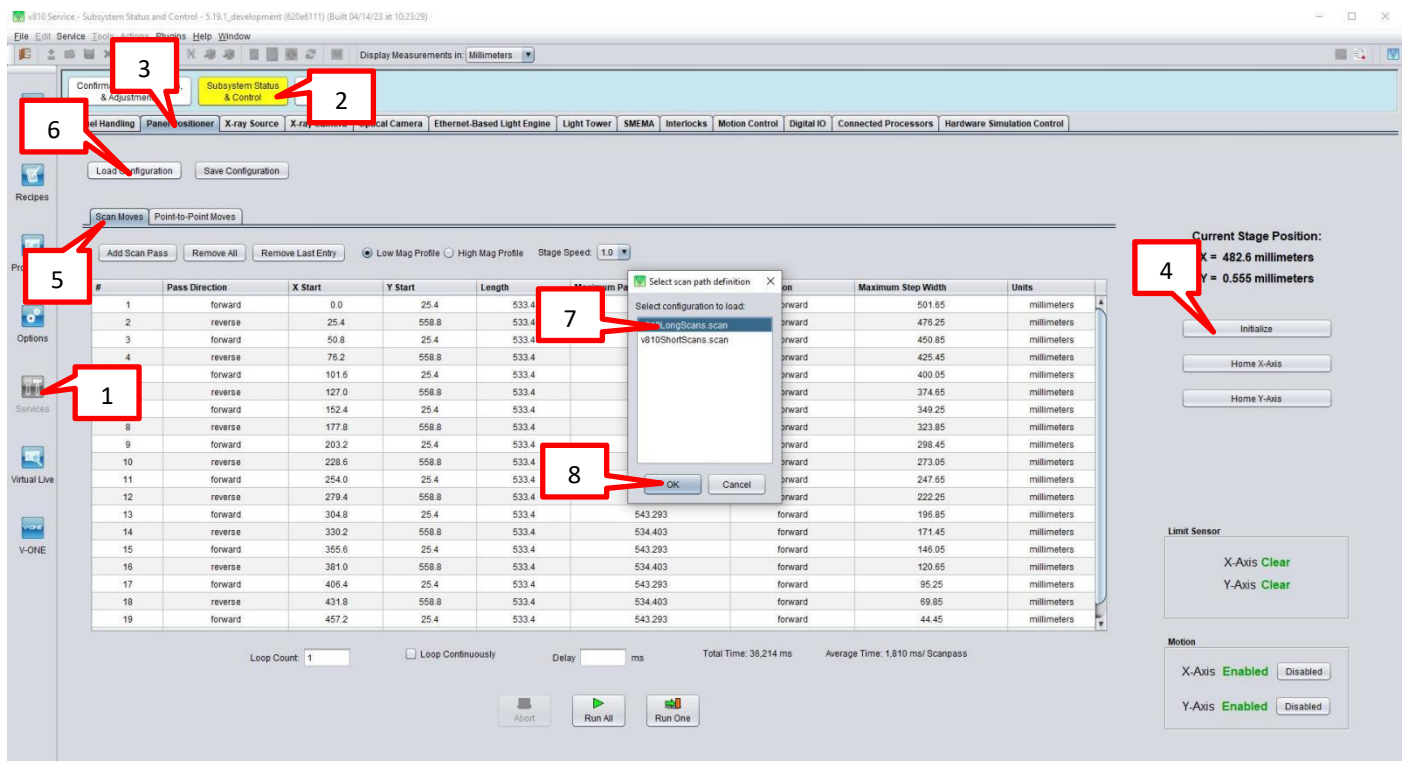


Figure 11 Load Long Scan Scan Moves

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14. At PANATERM application, click “MSR” to start measurement.

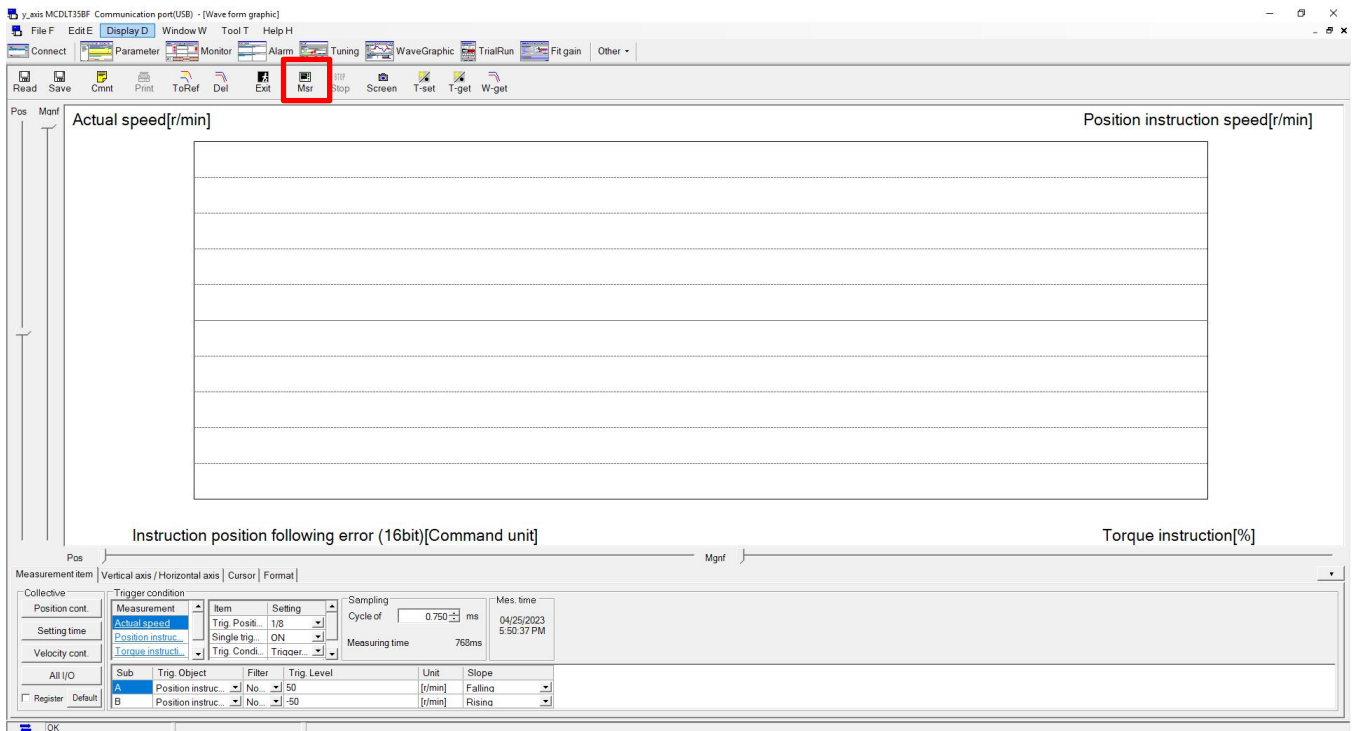


Figure 12 PANATERM Start Measurement

15. At V810 GUI, click “Run All”.

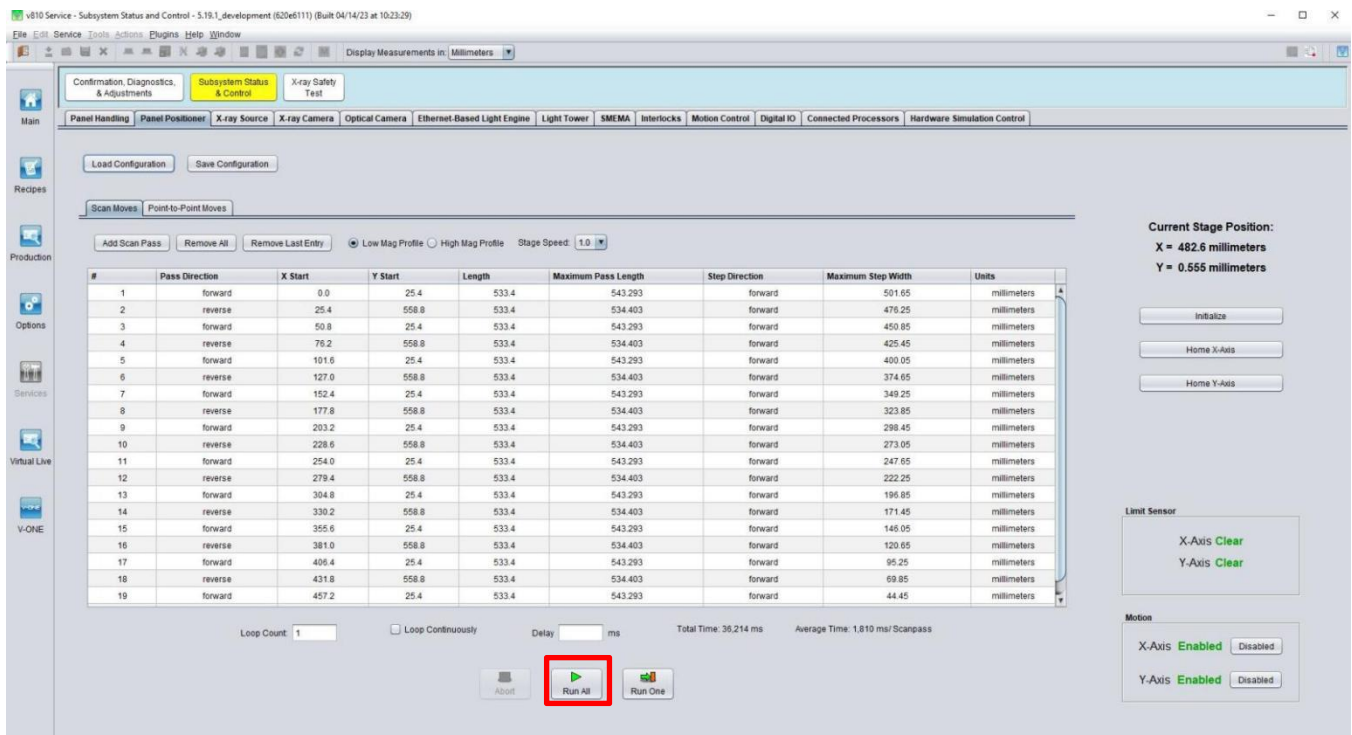


Figure 13 Start Long Scan

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16. After 30 seconds, the waveform will be transmitted from motion driver to PANATERM application.

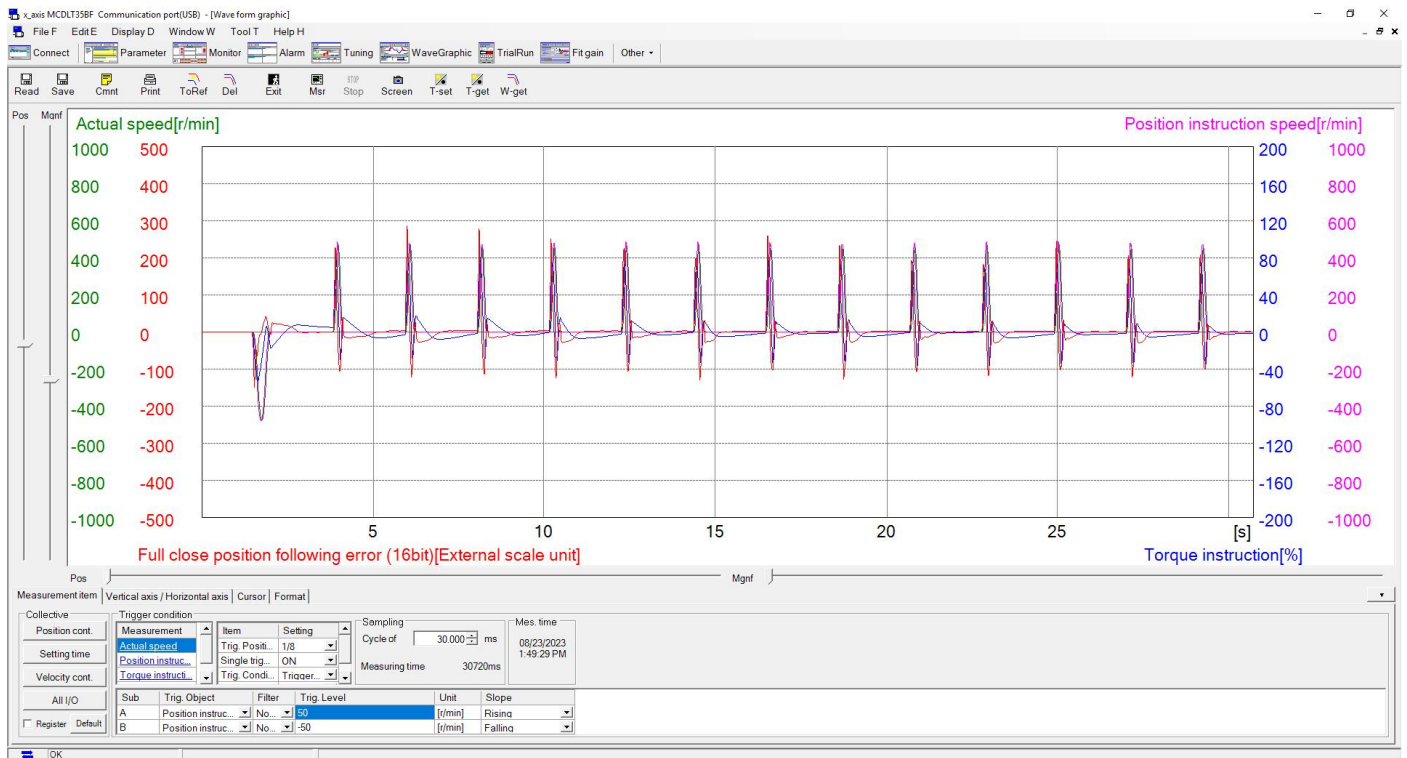


Figure 14 X axis Sample Motion Waveform

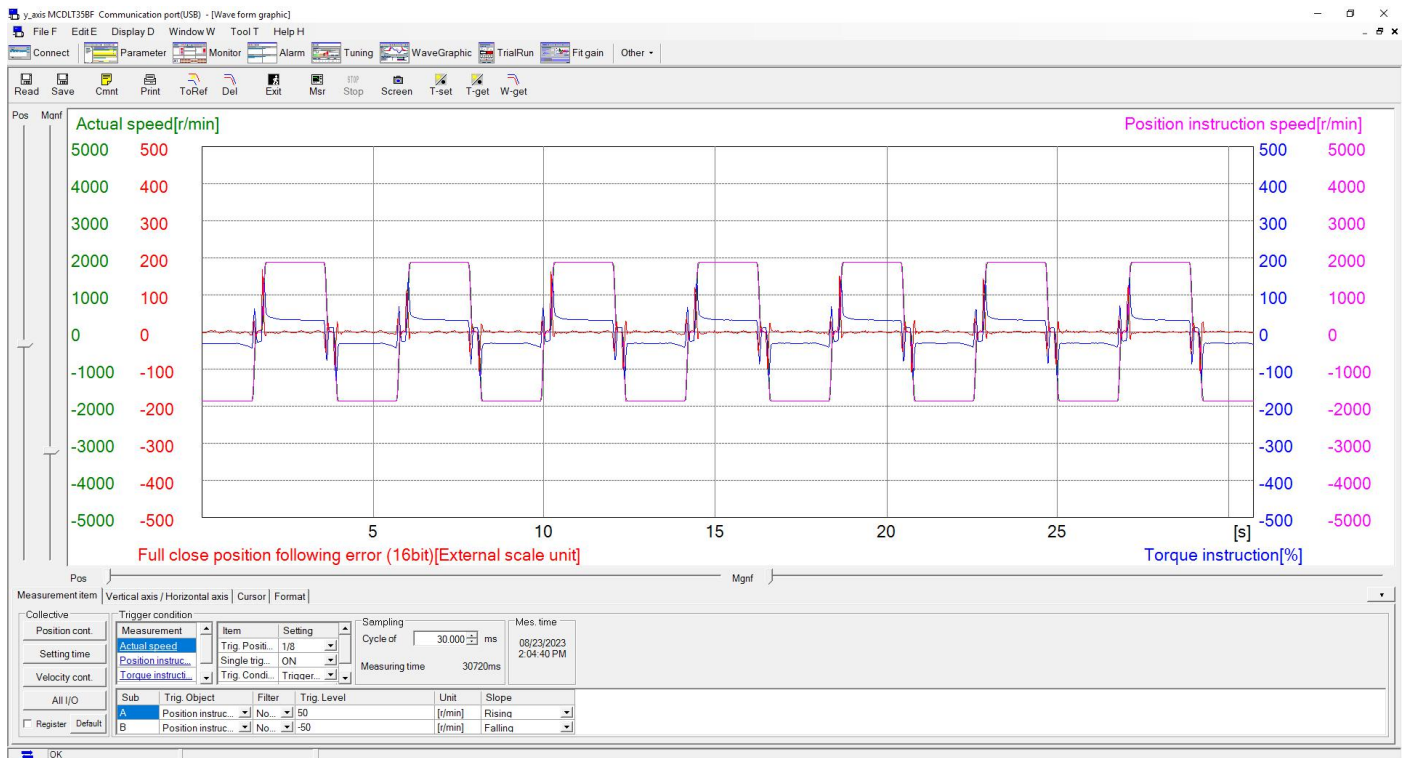


Figure 15 Y axis Sample Motion Waveform

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17. Click Save → Measuring result and parameter → OK to save the waveform.

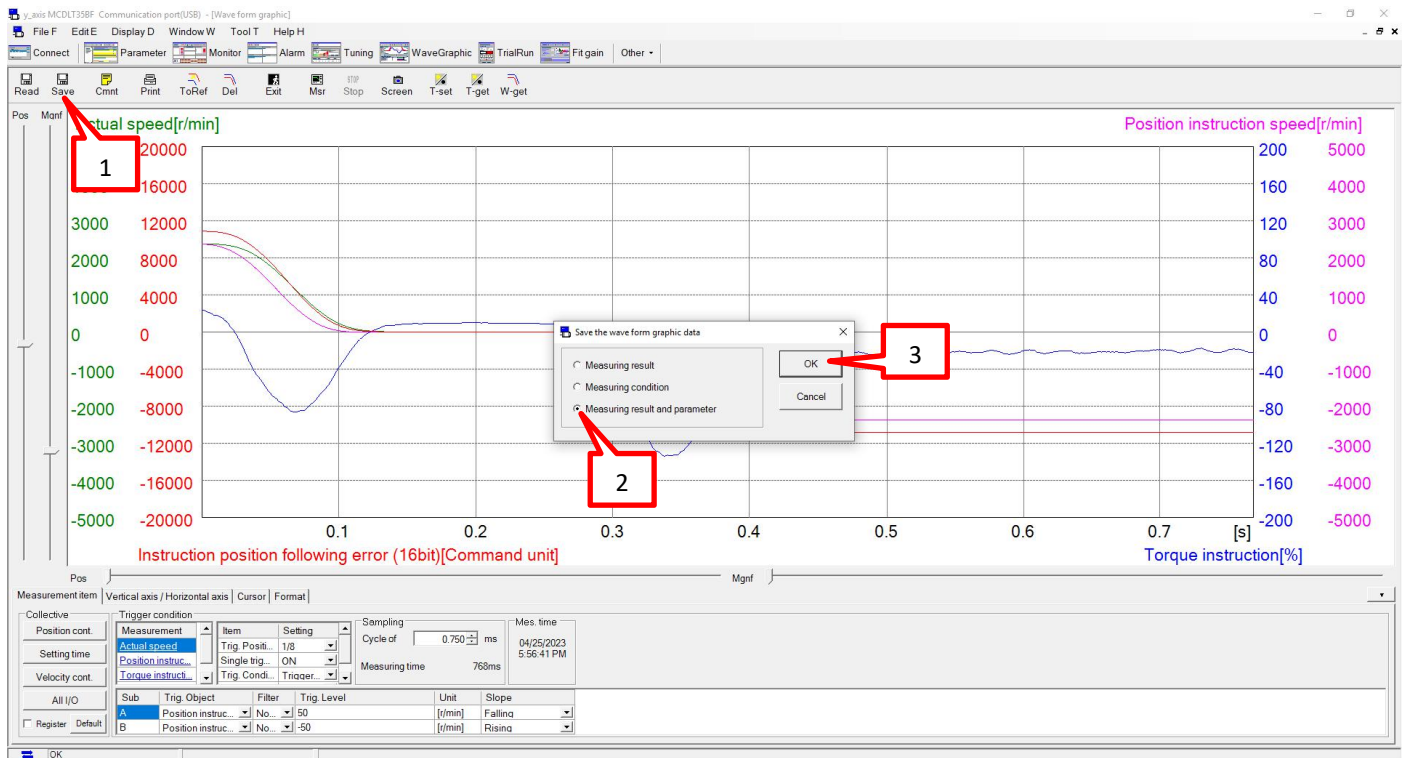


Figure 16 Save Waveform

18. Save the motion profile as *.wgp5 format with naming:

Axis	File Name
X-axis	[Date]_Xaxis
Y-axis	[Date]_Yaxis

Table 4 File Naming

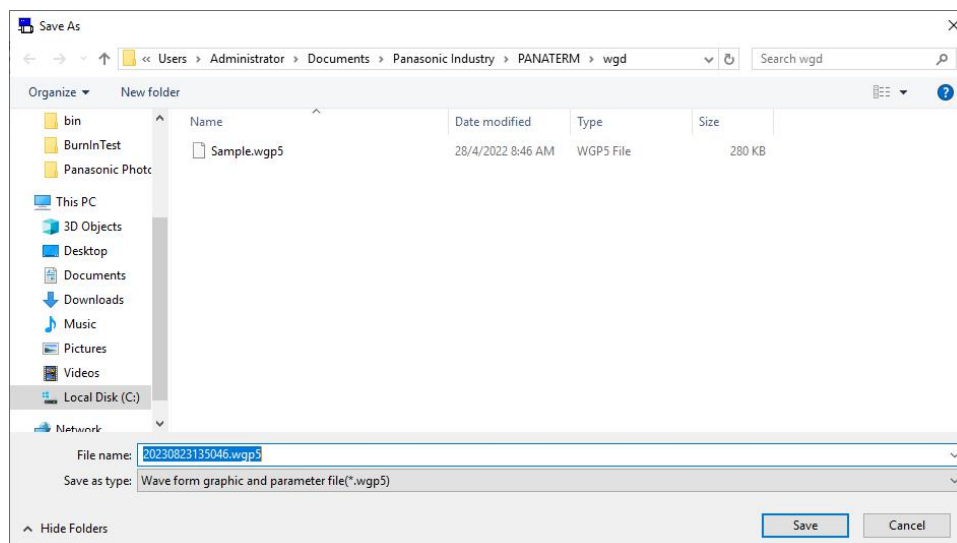


Figure 17 Save